

DoomSat: playing Doom through a real mission stack, with a System One and a System Two

Kevin Horton, 22 September 2026

Built with Claude Code, TypeSafe's *jev*, NASA F', Yamcs, NASA Open MCT and ViZDoom.

Repository: <https://github.com/Devonance/claude-jev-fprime-yamcs-openmct-doom>

1. What this is

Doom is a stress test dressed up as a toy: interactive (uplink latency shows up as a character that turns late), high rate (a frame every 100 ms has to be fragmented, framed, sent, deframed and reassembled) and self-validating (a dropped or mis-decoded frame is obvious to anyone watching). We put the game behind a real flight computer and played it from the ground through a real mission control system. The player is not a person: TypeSafe's **jev**, a System One model, classifies the situation every half second; **Claude Sonnet 5**, a System Two, nudges it once a minute and rewrites its questions after every attempt.

The result in one sentence: the stack works end to end under the load, every layer measured; the autonomous player explores, opens doors and dies honestly, but does not yet finish a level. The engineering result is the first part. The second part is documented as a failure with causes.

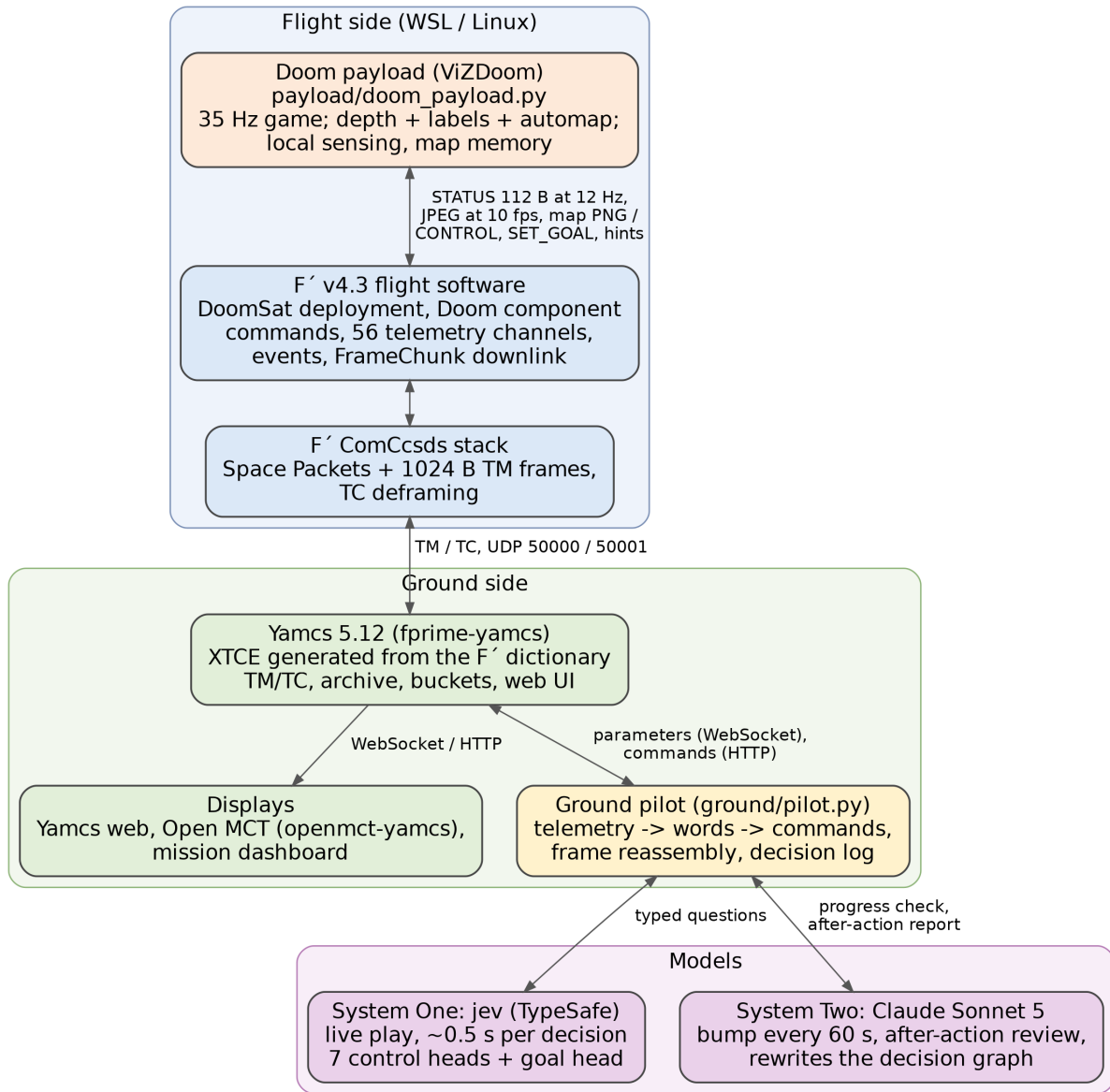


Figure 1: Architecture

2. The stack and how the data flows

Hop	What crosses it	Type and size	Rate
Doom → payload	screen, depth buffer, labels, automap (seen lines only), HUD variables	ViZDoom buffers, 640x480	35 Hz
payload → F' Doom component	STATUS record	112-byte big-endian struct, 56 fields	12 Hz
payload → F' Doom component	JPEG frame; map PNG every 5 s	~8 KB; ~2-6 KB	10 fps
F' component → com queue	56 telemetry channels; FrameChunk records	tlmWrite; 960-byte chunks as telemetry packets	12 Hz; ~90 chunks/s
F' ComCcsds → Yamcs	Space Packets in fixed TM frames	APID 1/2/3, 1024-byte frames, UDP	continuous
Yamcs → displays	parameters, events, command history, image products	WebSocket, HTTP, bucket objects	as they arrive
Yamcs → pilot	56 parameter values	WebSocket subscription	12 Hz
pilot → jev	structured state + 6-8 typed questions	HTTPS JSON, ~3 KB	every ~0.5 s
jev → pilot	choice + probabilities per Choice head, probability per Noul head	JSON	~450 ms after the ask
pilot → Yamcs → F'	CONTROL(move, strafe, turn, fire, use, weapon), SET_GOAL, EXPLORE_HINT, RESET_GAME	F' commands, CCSDS TC, UDP	one per decision
pilot → Sonnet	progress check (ASCII map + walk); after-action report	text, ~4 KB; JSON, ~2 KB	every 60 s; per episode
Sonnet → pilot	bearing + hold time + goal; revised graph config	JSON schema replies	20-120 s later

Everything on the ground reads Yamcs: the pilot, Open MCT and the dashboard have no other source.

Open MCT reads the same Yamcs: the frame product as an imagery view with its thumbnail strip, and any parameter as a plot (here jev's per-decision latency over 30 minutes).

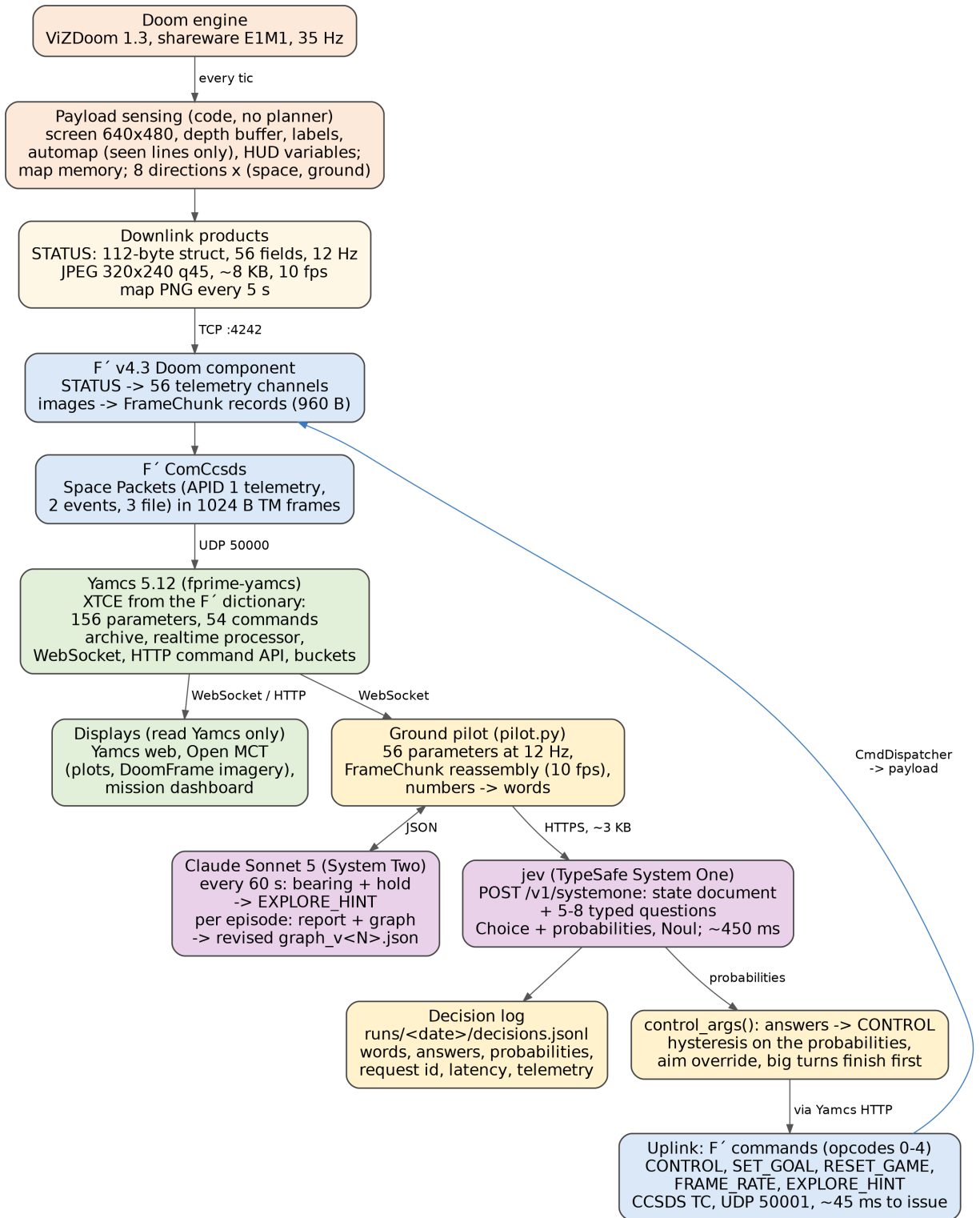


Figure 2: Data flow

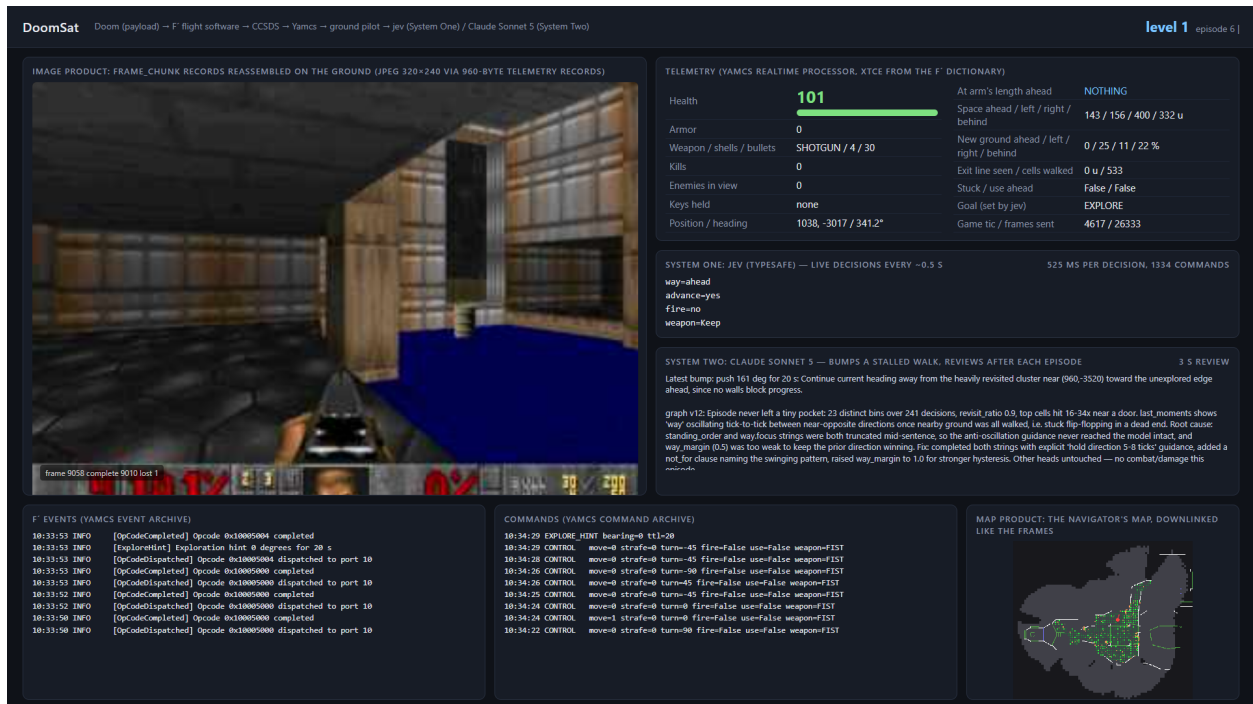


Figure 3: Dashboard

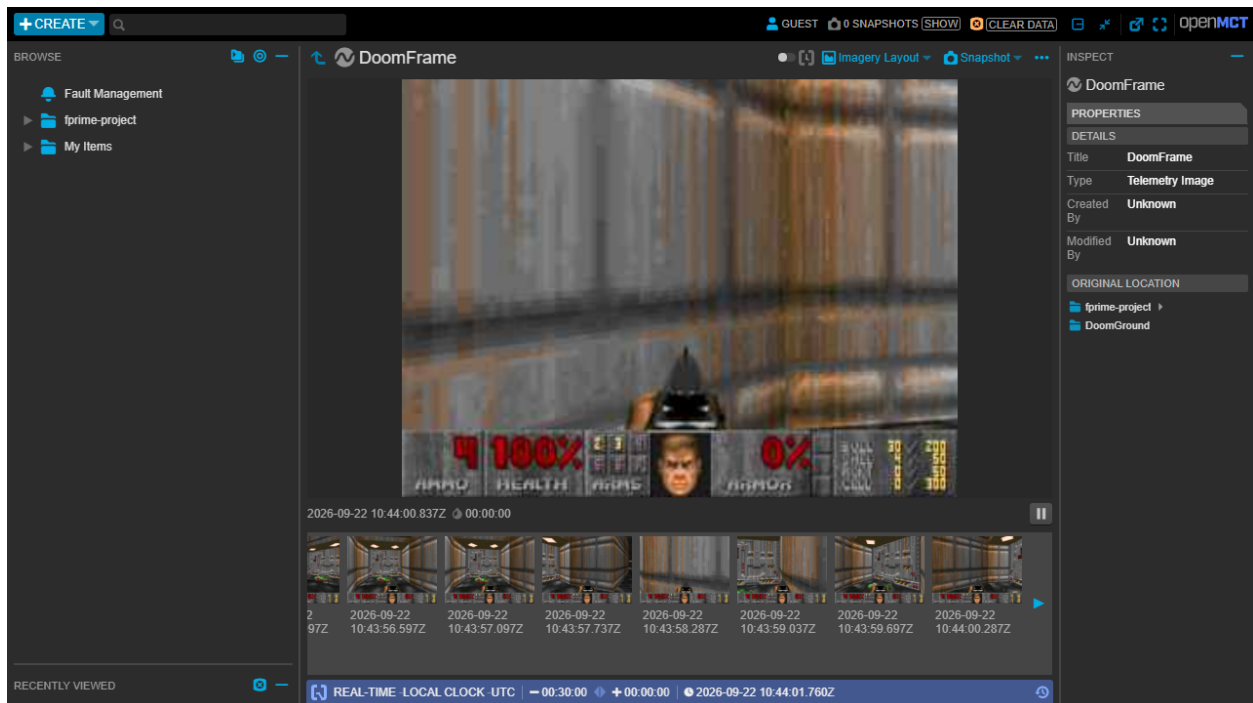


Figure 4: Open MCT imagery

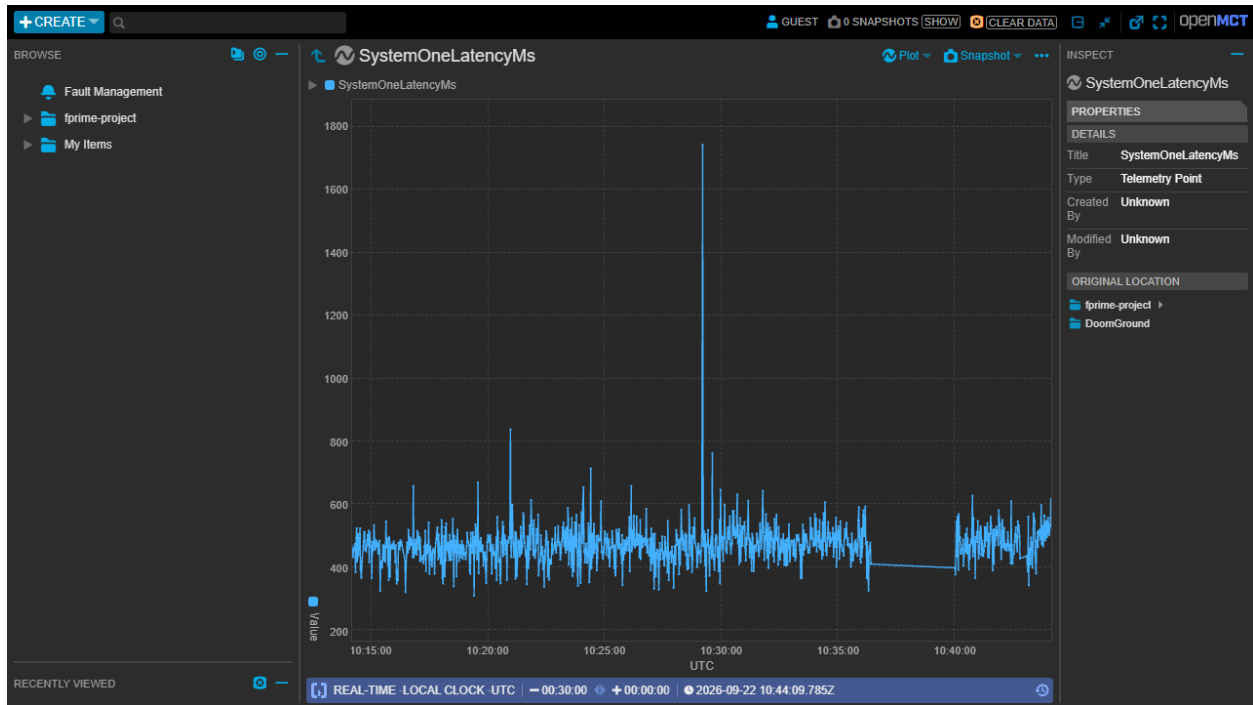


Figure 5: Open MCT plot

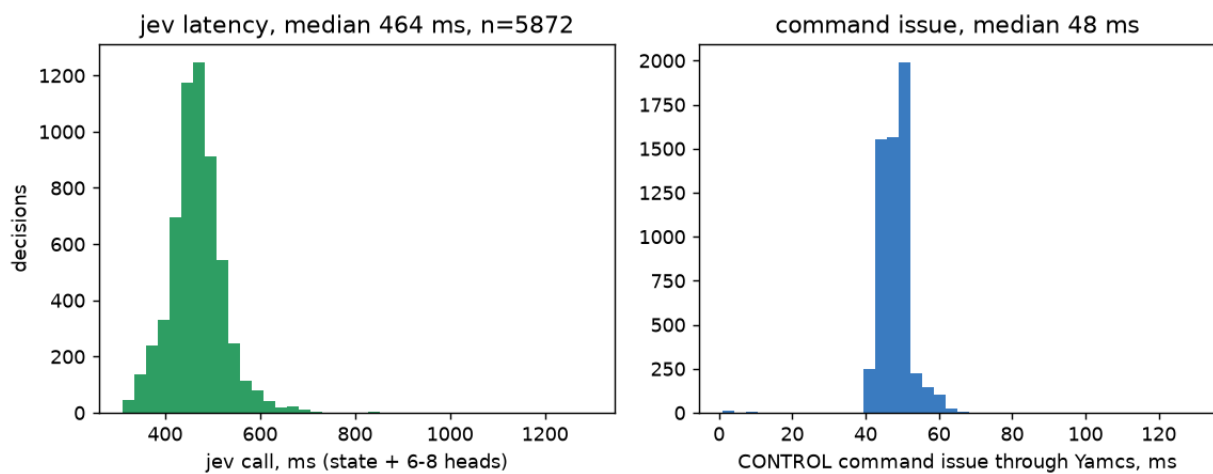


Figure 6: Latency

3. Measured, not asserted

Quantity	Value
XTCE from the F' dictionary	156 parameters, 54 commands, all Doom channels decode
Frame path	10 fps, 320x240 JPEG q45, ~8 KB, 9 chunks per frame, about 1 incomplete frame per 1000
Telemetry	STATUS at 12 Hz, 112 bytes, parsed into 56 channels
jev decision	~450 ms median per call including the Yamcs hop, 6-8 heads per call
Command issue	~45 ms through the Yamcs HTTP API
Turn execution	onboard setpoint, 6 degrees per tic; a 90-degree turn lands in ~0.45 s
Sonnet bump	20-60 s per call, one per minute
Sonnet after-action review	100-140 s per call, ~\$0.3, one per episode; graph revisions v2..v9 in one afternoon
Yamcs bucket	hard limit of 1000 objects; frames written into a ring of 20 names

Integration findings that cost real time and are worth keeping:

1. F' `string` telemetry is length-prefixed but `fprime-xtce` emits a fixed-size string type; Yamcs rejects every packet carrying one. Use enums.
2. Opaque byte arrays need the `!binary` annotation (`fprime-xtce` PR #8); the whole-struct form rejects array members.
3. `FW_COM_BUFFER_MAX_SIZE` must be raised (512 → 1000) through a `CONFIGURATION_OVERRIDES` module, not `settings.ini`.
4. The base rate group must tick faster than 1 Hz or the aggregator holds the last chunk of a frame.
5. Yamcs delivers F' booleans as the strings "True"/"False".
6. WSL processes launched from a `ws1 bash -c` call die with it unless started with `setuid -f`.
7. The `openmct-yamcs` example configuration imports a demo layout bound to a Yamcs instance called `myproject`; on any other instance Open MCT shows "Error requesting telemetry data" until that import is removed (ours lands on the frame product instead).

4. What the player is allowed to see

Nothing from the level file. The character senses like a player:

Sense	Stands in for	Source
Depth buffer	where the walls are	ViZDoom; perpendicular distance, 7.16 map units per step; 0 is the sky

Sense	Stands in for	Source
Object labels	recognising monsters, pickups, keys, barrels	ViZDoom labels buffer, things in view only
Automap, seen lines only	the map a player sees on Tab	ZDoom NORMAL mode; walls, floor steps, ceiling changes (doors), locked doors in key colour, exit lines; only recoloured
HUD variables	health, armor, ammo, position, heading	ViZDoom game variables

Not used: whole-map or show-objects automap modes, trigger-line display, sector geometry from the game state, monster counts, item lists, cheats. The WAD statistics script is a developer tool for choosing levels and checking results; the payload never imports it.

The payload has no planner. It stamps the automap into a raster, sweeps the floor it has seen, counts where it has walked, and reports eight directions around the player (every 45 degrees): how far the map is open, and whether the ground that way is unexplored, new, walked before, or has a door on the way. It also reports what is at arm's length ahead (wall, door, exit switch, locked door, monster or barrel, or something the map does not show such as window bars), where an exit line, a key and pickups were seen, and whether the player is stuck. What it bumps into becomes a barrier for the rest of the level. The map of a level is kept across attempts; the game itself restarts.

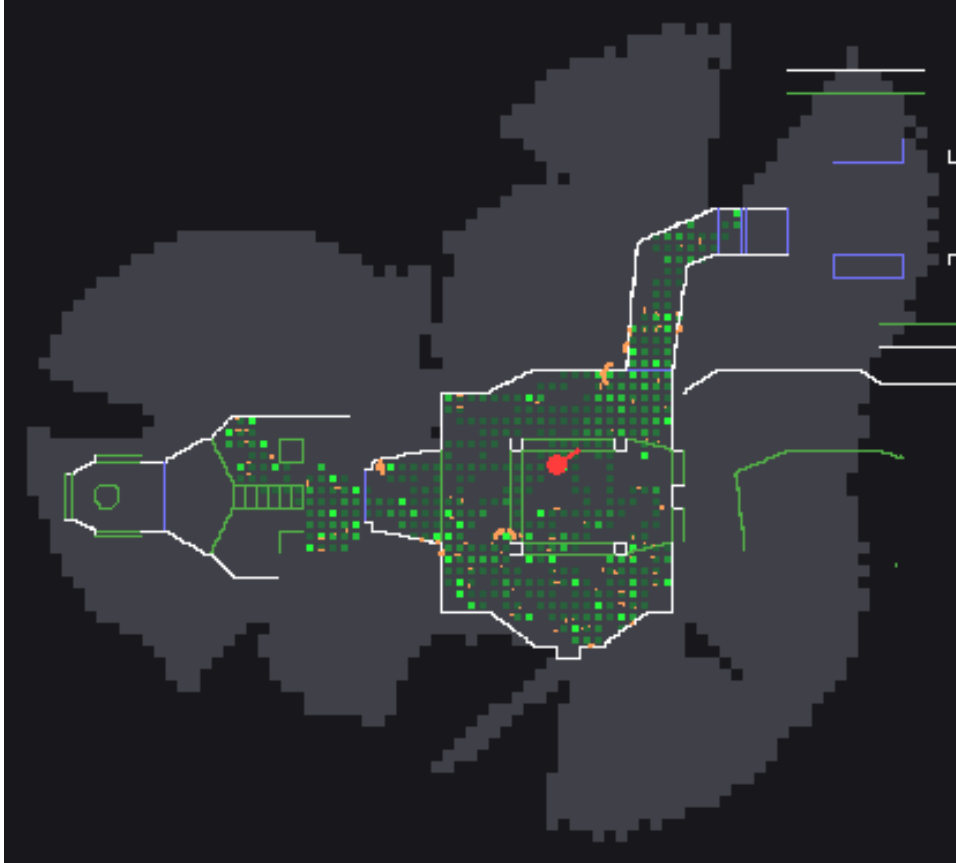


Figure 7: Map product

5. jev's decision graph

jev classifies a structured state document in one forward pass and returns, per question, an option with probabilities (Choice) or a yes-probability (Noul). It does not reason, plan or remember. The graph is therefore one narrow question per head, options with contrastive criteria (what / not_for / examples) that reference state fields, Nouns for yes/no heads, and code combining the answers:

Head	Type	Options	Decides
way	Choice	the open directions among ahead, ahead-left, left, behind-left, behind, behind-right, right, ahead-right, each with its own "now" (space, ground)	the direction: jev is the navigator
advance	Noul	yes / no	walk forward this tick
use	Noul	yes / no	press Use (asked with a door, the exit or a locked door at arm's length, or when stuck)
fire	Noul	yes / no	with a living enemy in the crosshair and ammunition
dodge	Choice	Carry on / Dodge left / right / back (open sides only)	evasion from a close enemy
turn	Choice	Hard left ... Hard right / Turn around	aim at a visible enemy
weapon	Choice	Keep / Pistol / Shotgun	ammunition scarcity
goal (every 8th tick)	Choice	Kill enemies / Restore health / Stock ammo / Add armor / Explore	priority

Code maps the answers onto one CONTROL command: the way becomes a turn setpoint of 0, 45, 90, 135 or 180 degrees (or a step backward when the way behind is open), advance becomes the move, and the option probabilities give hysteresis so the direction only changes when jev is clearly surer of the new one. A large turn is allowed to finish before the next question is asked.

System Two on two clocks: every minute Sonnet gets seconds into the attempt, cells gained, position, the walk's most visited spots and the map product as text, and answers with a bearing, a hold time and a goal (sent as `EXPLORE_HINT` and `SET_GOAL`). After every episode (death, level finished, or the 3-minute budget spent and the game reset) it gets a ~2 KB report (outcome, health, damage, cells, distance, stuck ticks, per-head answer distributions with confidence, the walk, the last eight decisions in words) plus the current graph, and returns a revised graph that code validates and versions.

The graph as a table, in the form of the rover demo's decision graphs: what code turns into words, the typed question jev is asked, when, and the rule that consumes the answer. Nothing slower than jev is in the live loop.

Table 5: jev’s decision graph: what code turns into words, the typed question, when it is asked, and the rule that consumes the answer. Measured on 22 Sep 2026: 5,872 jev decisions through the stack, about 450-520 ms per call with 5-8 heads, about 45 ms to issue the command; Sonnet: a bump per minute and 14 graph versions in the day.

Head and the words it sees	Typed question	When asked	Rule that consumes the answer
1. way (the navigator) eight directions around the player, 45 deg apart, each in words: space (blocked / tight / open / long) and ground (unexplored / new / partly walked / walked before / a door on the way at N units); what is at arm’s length ahead; where the exit and a key were seen; the goal word	Choice over the directions that are not blocked (2 to 8 options); every option carries what / not_for / examples criteria plus its own ‘now’	every tick (~0.5 s), one request with all heads	turn setpoint by direction: 0 / 45 / 90 / 135 / 180 deg (right = negative); the previous way is kept unless $P(\text{new}) \geq P(\text{previous}) + \text{way_margin}$ (hysteresis on the probabilities); ‘behind’ with open ground behind = one step back instead of a U-turn; a turn of 45 deg or more is issued alone and finishes before the next ask
2. advance the same state document; ‘ahead’ carries space, ground and what is at arm’s length	Noul: walk forward this tick?	every tick	move = 1 if $P(\text{yes}) \geq 0.5$; suppressed while an enemy is in view and the aim is off by more than aligned_deg (35)
3. use at arm’s length ahead: door / exit switch / locked door (with the key colour) / wall / nothing; stuck flag	Noul: press Use now?	only when a door, the exit or a locked door is at arm’s length, or the player is stuck	use = 1 if $P(\text{yes}) \geq 0.5$; the payload marks a door that stays shut after its presses as a barrier for 120 s
4. fire nearest enemy: bearing in words (in the crosshair / left / right / behind), distance band, health, ammo	Noul: shoot now?	every tick	fire = 1 if $P(\text{yes}) \geq 0.5$ and the enemy is within fire_range (450 u) and crosshair_deg (8); never without ammo
5. dodge nearest enemy bearing and distance band; which sides are open	Choice: Carry on / Dodge left / Dodge right / Dodge back (only the open sides are offered)	whenever an enemy is in view	strafe = -1 / +1, or move = -1 for ‘Dodge back’; danger_dist (180 u) gates the question’s ‘close’ word
6. turn (the aim) nearest enemy bearing in words	Choice: Hard left / Left / Fine left / Hold / Fine right / Right / Hard right / Turn around	whenever an enemy is in view	overrides the way’s turn with the aim table (turn_deg: 60 / 25 / 8 / 0 / -8 / -25 / -60 / 150 deg); a person does not keep exploring under fire
7. weapon weapon in hand, shells, bullets, enemy distance band	Choice: Keep / Pistol / Shotgun	every tick	weapon field of CONTROL; ‘Keep’ sends no switch

Head and the words it sees	Typed question	When asked	Rule that consumes the answer
8. goal health and armor bands, ammo, kills, enemies in view, cells walked, exit seen, pickups seen with bearings	Choice: Explore / Kill enemies / Restore health / Stock ammo / Add armor	every goal_every ticks (8) or when the picture changes	SET_GOAL command; the goal word enters the next state documents and the payload's pickup targeting
System Two: bump (Claude Sonnet 5) seconds into the attempt, cells gained, position, most visited spots, the map product as text	not jev: a JSON-schema reply with bearing, hold time and an optional goal	every 60 s	EXPLORE_HINT(bearing, ttl): the payload steers 'ahead' toward the bearing while it lasts; jev still answers every head
System Two: after-action (Claude Sonnet 5) the episode report: outcome, health, damage, cells, distance, stuck ticks, answer distributions per head, the walk, the last eight decisions in words, plus the current graph	not jev: a revised graph (criteria text, thresholds, way_margin, goal_every) with a rationale	after every episode: death, level done, or the 180 s budget	code validates (fixed option names, known placeholders, clamped thresholds), stores graph_v<N>.json, and jev plays the next attempt with it

One real decision from the last run, end to end, with the numbers Yamcs delivered, the words code made from them, jev's probabilities and the command that went up:

Yamcs parameters (12 Hz), this tick

CLEAR ahead 116, ahead-left 48, left 28, behind-left 20, behind 24, behind-right 248, right 212, ahead-right 120 (units)
NEW ahead 0, ahead-left 100, left 0, behind-left 0, behind 0, behind-right 14, right 50, ahead-right 33
AHEAD_KIND NOTHING, EXIT_DIST 0, STUCK false, ENEMY_COUNT 0, HEALTH 100, ARMOR 0, KILLS 0
POS 1347,-2725, ANGLE 216, EXPLORED_CELLS 544, LEVEL 1, KEYS 0

↓ *telemetry → words (bands, novelty, direction names); no number reaches jev*

State document (code, no model)

ahead: open, walked before, nothing near
ahead-left: tight, new
left / behind-left / behind: blocked, walked before
behind-right: long (a passage or a big room), walked before
right: open, partly walked; ahead-right: open, partly walked
exit: not seen; enemy: none in view; health: full; armor: none; goal word: Explore

↓ *HTTPS POST /v1/systemone: the state plus 5 typed questions with their criteria*

jev, one request, 469 ms (req_01a0c8b54b2c73dd8ec79b57a6f3a1b6)

way (Choice over the 5 open directions): ahead-left 0.82, ahead 0.14, right 0.03, ahead-right 0.01, behind-right 0.00
advance (Noul): yes 0.75
fire (Noul): no 0.98
weapon (Choice): Keep 0.82, Shotgun 0.18, Pistol 0.00
goal (Choice, this was a goal tick): Add armor 0.86, Explore 0.14

↓ *choices with probabilities, Noul probabilities*

Code rules

way: the previous way 'ahead' holds 0.14; the new one 0.82 $\geq$ 0.14 + way_margin, so switch; ahead-left = +45 deg
a turn of 45 deg or more is issued alone: move = 0 this tick, walking resumes when the turn has landed
fire 0.98 no → fire = 0; use was not asked (nothing at arm's length) → use = 0
weapon Keep → no switch
goal Add armor → SET_GOAL(ADD_ARMOR) on the goal tick; the word 'Add armor' enters the next documents

↓ *one command per decision*

CONTROL(move=0, strafe=0, turn=+45, fire=0, use=0, weapon=no change)

Yamcs HTTP → CCSDS TC → UDP → F CmdDispatcher → Doom component → payload turn setpoint (6 deg per tic)
issued in 50 ms; F events: OpCodeDispatched, OpCodeCompleted

Figure 8: One real decision end to end (episode 7, graph v14, 22 September 2026 10:41:59 UTC): the numbers Yamcs delivered, the words code made from them, jev's probabilities, the code rules and the command that went up.

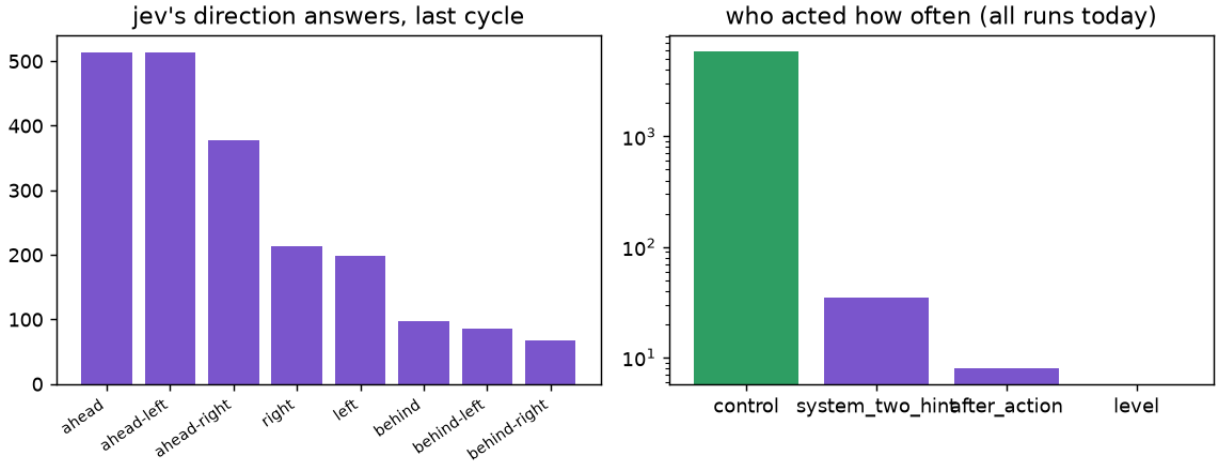


Figure 9: Answers

6. Results: the stack held, the player did not finish

Runs of 22 September 2026 on shareware E1M1, jev live through the whole stack, Sonnet 5 as System Two. Each cycle is run / review / edit / run. Cells are 32-unit map cells the character has stood in.

Cycle	Change	Decisions	Cells	Stuck	Reached	Notes
planner v1	frontier planner over the automap	687	121	60	start room	walked into a fake door repeatedly; turns alternated hard left/right
planner + steer	jev picks the way out when blocked	202	147	0	start room	the planner still sealed its own cell
local 1	no planner; four directions, prose criteria	124	56	1	start room	"Turn around" 62 times: sides read blocked in every corridor
local 2	eight directions, structured criteria, Nouis	151	91	4	start + north room	balanced directions, advance yes 60%
local 3	rays stop at unseen ground; big turns finish first	155	103	1	start, north, west	first clean run
local 4	unseen ground is "unexplored", the strongest pull	155	126	0	west wing (a dead end)	4.8k units walked
local 5	map kept across attempts	250	246	7	whole west + north	the second attempt doubles the map

Cycle	Change	Decisions	Cells	Stuck	Reached	Notes
local 6	doors on the way in each direction's words	295	275	3	through the silver door into the big room	first door opened (23 Use presses); died to the first monsters, 0 kills
local 7	aim/dodge whenever an enemy is visible; blocked directions are not "new"; map kept over 7 attempts	3170 total	559	11	start, north, west, the door, first kill	Sonnet graph v9..v14; the loop kept re-walking known rooms

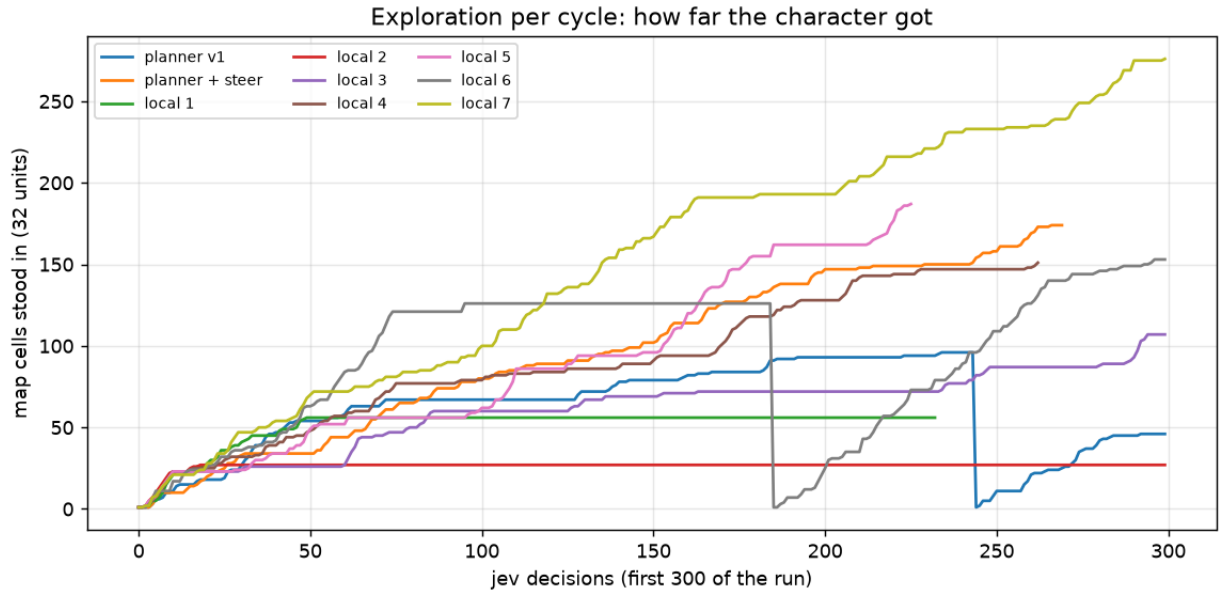


Figure 10: Exploration per cycle

Sonnet's revisions were specific and correct about what they saw: "Hold answered 78%, aligned_deg too strict, loosened to 50"; "the goal flipped every 4 ticks between explore and add armor, raised to 8"; "the standing order was truncated mid-word" (it was: a validator limit, since fixed); "way flipped nearly every tick, raised the hysteresis margin". None of them could fix what the words did not carry.

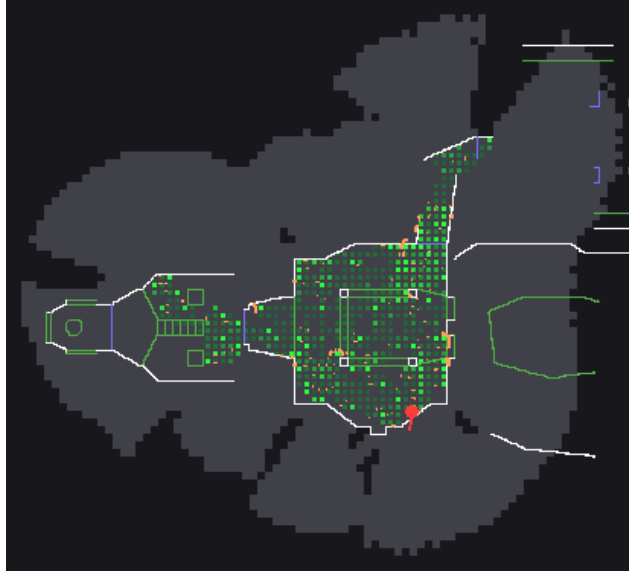


Figure 11: Explored map at the end of the day

7. Why the player did not finish, and what to do

1. **Sensing errors, not decisions.** Most of the day went into finding that jev's inputs were wrong: depth scale and geometry, the sky as distance zero, the crosshair and player arrow drawn into sensor buffers, barred windows the automap never draws, barrier marks inside the character, a blocked direction reading "new ground". jev answered every wrong input correctly. Each fix moved the character further.
2. **A planner hid the errors.** The first navigator turned them into confident wrong routes. Removing it and making jev the navigator (eight directions in words) was the right move and should have been the first design.
3. **The graph was written for an LLM at first.** Compound prose criteria, no Nouns, probabilities ignored. The structured graph came late; it is the version to build on.
4. **Coverage per attempt is too low.** 200-300 decisions in 3 minutes cover 100-250 cells; E1M1 needs a walk of several thousand units through two doors. With the map kept across attempts the loop does make progress, but it re-walks known rooms too often; a Score per direction ("how promising for finding the exit") combined in code is the next thing to try.
5. **What a wall is cannot be read from the automap alone.** Doors and exit lines are coloured; switches, lifts and signs are not. A small vision classifier over the screen patch at arm's length (door / switch / exit sign / wall / monster / pickup), or jev fed with a few numbers from that patch (dominant colours, edge density, whether it tiles, its height), is the honest replacement. This is where a vision decision model would slot in.
6. **Combat is barely tested.** The aim and dodge heads now fire whenever an enemy is visible. The first fight of the day was lost with no shots fired because the goal head had not switched to fighting; in the last attempt of the day the character answered fire 3 times, made its first kill and came out at 13 health with one shell left. One fight is not a result.
7. **Height is invisible.** The range camera is at eye level; drops and platforms are learned by bumping. A second depth band below the horizon would give "a drop ahead" cheaply.

The handoff for the next pass, with the full list of what made us stuck and the research leads, is docs/HANDOFF.md.

8. Sources

- NASA F' v4.3.0: <https://github.com/nasa/fprime>
- fprime-yamcs 0.2.1: <https://github.com/fprime-community/fprime-yamcs>; fprime-xtce (binary annotation branch): <https://github.com/FarkasJoseph/fprime-xtce/tree/feature/binary-annotation-combined>
- Yamcs 5.12.8: <https://github.com/yamcs/yamcs>; yamcs-client 2.1: <https://github.com/yamcs/python-yamcs-client>
- NASA Open MCT: <https://github.com/nasa/openmct>; openmct-yamcs: <https://github.com/akhenry/openmct-yamcs>
- TypeSafe jev and System One docs: <https://docs.typesafe.ai/introduction>, <https://docs.typesafe.ai/primitives/advanced>, <https://docs.typesafe.ai/concepts/how-to-build-with-system-one>, <https://docs.typesafe.ai/sdk/python/api>
- Claude Sonnet 5 through Claude Code: <https://code.claude.com/docs>
- ViZDoom 1.3.0: <https://vizdoom.farama.org>, <https://github.com/Farama-Foundation/ViZDoom> (ZDoom automap source: https://github.com/Farama-Foundation/ViZDoom/blob/master/src/vizdoom/src/am_map.cpp)
- Shareware Doom IWAD (Debian doom-wad-shareware); Freedoom: <https://freedoom.github.io>
- The System One demo this graph descends from (its Doom demo reads the WAD; ours does not): <https://github.com/sgoeddecke/system-one>
- Prior LLM-plays-Doom work: <https://adriandewynter.substack.com/p/will-gpt-4-and-5-run-doom>
- Graphviz (diagrams), Playwright + Chrome (screenshots, recording), tectonic (PDF), imageio-ffmpeg (video)